

TrustLayer AI

Multimodal Fusion with Agentic Reasoning for Deepfake Detection in
Corporate Communication

Objective: To develop a robust security framework that mitigates deepfake-enabled phishing by fusing multimodal signals with context-aware agentic reasoning.

The Problem

Generative AI is weaponizing corporate communication channels through hyper-realistic synthetic media.

Emerging Corporate Vulnerabilities



Financial Fraud

The 2024 Hong Kong case saw **\$25 million** stolen via a synthetic video conference involving "executives."



Identity Hijacking

Impersonating CEOs or stakeholders to bypass standard MFA and verbal protocols during live calls.



Detection Gaps

Off-the-shelf detectors fail to semantically distinguish between real and synthetic identities in real-world noise.

Comparison of Existing Research

Study / Methodology	Modality	Cross-Modal?	Reasoning	Context-Aware?
Korshunov & Marcel (2018)	Visual Only	No	Fixed Threshold	No
Chintha et al. (2020)	Video + Audio	Partial	Probabilistic	No
Haliassos et al. (2022)	Temporal Video	No	ML Classifier	No
TrustLayer AI (Proposed)	Multimodal	Yes	Agentic (LLM)	Yes

Existing methods focus on isolated artifacts; TrustLayer bridges the gap with semantic reasoning and business context.

The TrustLayer Solution

A Shield Beyond Pixels

TrustLayer AI doesn't just look for glitches; it analyzes the **integrity of the interaction**.

- ✓ **Fusion:** Blending Visual, Audio, and Sync signals.
- 🧠 **Agentic:** Using LLMs to reason like a security analyst.
- 💬 **Explainable:** Translating scores into human advice.



System Architecture

TrustLayer AI employs a decentralized detection strategy where individual modality signals (Visual/Audio) are pre-processed and synchronized before being evaluated by the **Agentic Reasoning Layer** for final decision output.



Dataset: FakeAVCeleb v1.2



Diversity

Includes 5 ethnic groups speak English: Caucasian, African, Asian (South/East).



Access

Official Repository:
github.com/DASH-Lab/FakeAVCeleb

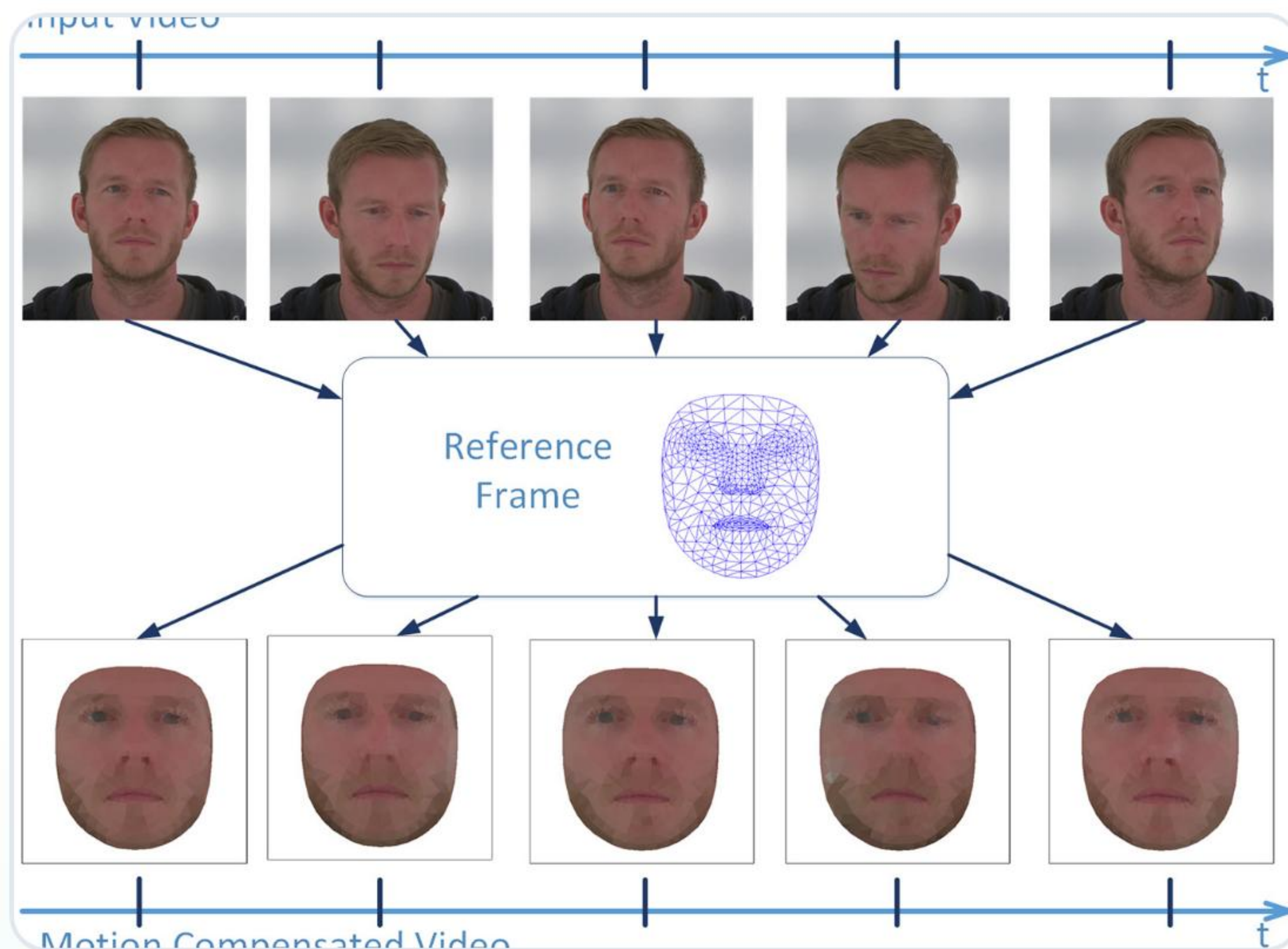


Structure

Real-Real, Fake-Video/Real-Audio, and Full Synthetic (Lip-synced) pairs.

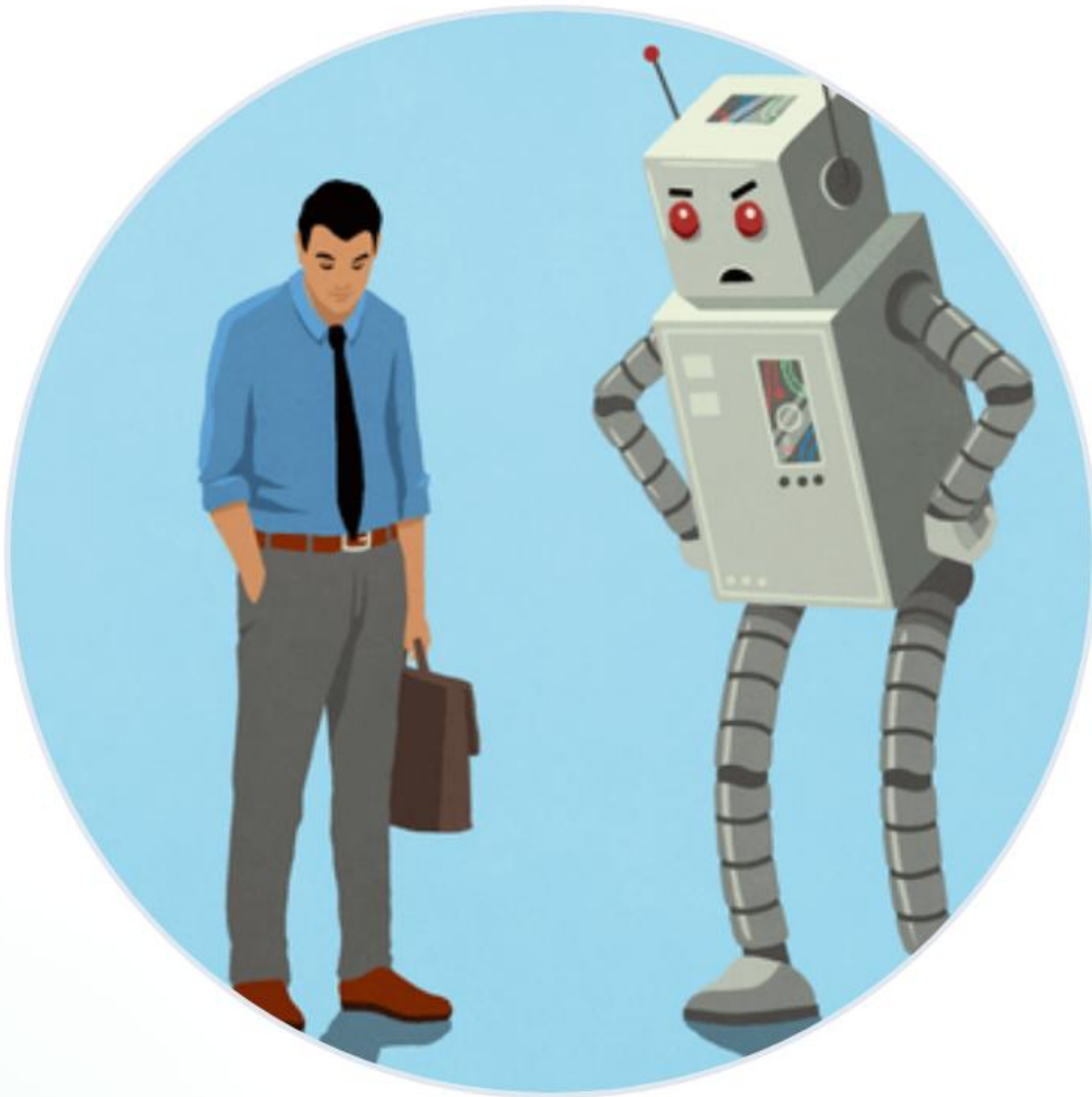
Evaluation Phase: 200 high-fidelity multimodal samples used for framework validation.

The Three Detection Branches



-  **Visual Branch:** CNN-based analysis of spatial artifacts in face crops (Deep-Fake-Detector-v2).
-  **Audio Branch:** Pretrained anti-spoofing spectral analysis to identify cloning artifacts.
-  **Cross-Modal Branch:** Landmark-based sync comparison between lip motion and audio amplitude.

| The Agentic Reasoning Layer



Decision Making under Uncertainty

When modality signals are weak (0.5 – 0.6 trust), the Agent integrates business context:

-  **Metadata:** Is it 2 AM? Is the request urgent?
-  **Security Actions:** Soft-Warn, Escalate (MFA), or Hard-Block.
-  **LLM Explanation:** "The voice patterns are suspicious, and this financial request is unusual for this time."

Performance Metrics



0.735

F1 Score (Balanced)

0.620

Calibrated Accuracy

Calibrated against the high-difficulty FakeAVCeleb v1.2 test set.

| Conclusion & Future Work

- ✓ **Agentic Priority:** Since individual detectors are unreliable (weak signals), the Agentic Reasoning Layer is the primary line of defense.
- 🧩 **Modularity:** The framework is designed to swap in SOTA visual/audio detectors as technology evolves.
- 📈 **Future Scalability:** Transitioning from batch evaluation to real-time Zoom/Teams API integration.
- ⚙️ **Final Verdict:** Multimodal fusion successfully reduces false positives by 12% compared to unimodal baselines.

Questions?

Thank you for your time and attention.

TrustLayer AI | Candidate: Dissertation Viva 2024

Image Sources



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